

Sri Lanka Institute of Information Technology



Web – based School Information Management System

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SE2030 – Software Engineering

07-August-2026

1. Introduction

1.1 Project Overview

The Web-Based School Information Management System (SIMS) is a centralized Java-based web application developed to improve the management of academic and administrative activities of Wycherley International school Gampaha. The proposed system integrates various school operations into a single platform, allowing authorized users to efficiently manage student records, teacher information, attendance, timetables, examinations, and fee payments. Instead of relying on multiple manual processes or disconnected software, the system provides one centralized database that ensures consistent, secure, and organized information management.

Modern schools generate a large amount of academic and administrative information every day. Managing these records manually is time-consuming and increases the possibility of errors. By introducing a web-based solution, the school can improve productivity, enhance communication among stakeholders, and provide quick access to accurate information whenever required.

1.2 Problem Background

Many schools still rely on traditional methods such as paper files, handwritten registers, and spreadsheets to maintain student and staff information. These manual approaches often create several operational challenges.

Student records, attendance details, examination marks, and fee information may be scattered across multiple files, making it difficult to retrieve accurate information quickly. Preparing reports manually requires considerable time and effort, resulting in delays when administrators need important information for decision-making. Furthermore, parents and students have limited access to academic updates, while duplicate or incorrect records may occur because information is entered multiple times in different locations. These issues reduce the efficiency of school administration and increase the workload of teachers and administrative staff.

1.3 Proposed Solution

To overcome these challenges, this project proposes a Web-Based School Information Management System that provides a centralized platform for managing all major school activities. The system allows different categories of users—including administrators, principals, teachers, students, parents, and the Head of Academic—to access features according to their responsibilities through role-based authentication.

The proposed system stores all information within a centralized database, reducing data duplication while improving consistency and accuracy. Each functional module supports Create, Read, Update, and Delete (CRUD) operations, enabling authorized users to efficiently

manage records. The system also provides faster report generation, improved data accessibility, and better communication between all stakeholders.

1.4 Project Objectives

The primary objectives of the proposed system are as follows:

- **Objective 1 – Centralize School Records**

The system will store student, teacher, attendance, timetable, examination, and fee information within a single centralized database. This eliminates scattered records and provides easier access to information.

- **Objective 2 – Support Different School Roles**

The system provides different access privileges for Principals, Administrators, Teachers, Students, Parents, and the Head of Academic. Each user can access only the information relevant to their responsibilities, ensuring security and privacy.

- **Objective 3 – Provide Complete CRUD Operations**

Each major function allows authorized users to create, read, update, and delete or deactivate records where appropriate. These operations simplify day-to-day school management activities and improve data accuracy.

- **Objective 4 – Improve Administrative Decision Making**

The proposed system generates accurate reports and summaries that help school management monitor academic performance, attendance, fee collection, and other administrative activities more efficiently.

1.5 Target Users

The proposed system is designed for multiple stakeholders within the school environment. They are:

- Principal
- School Administrator
- Teachers
- Students
- Parents
- Head of Academic

Each user will have role-based access to the system, ensuring that they can only perform the functions relevant to their responsibilities. This approach improves security while maintaining efficient collaboration among different user groups.

1.6 Project Scope

The scope of the proposed system includes the development of six major functions:

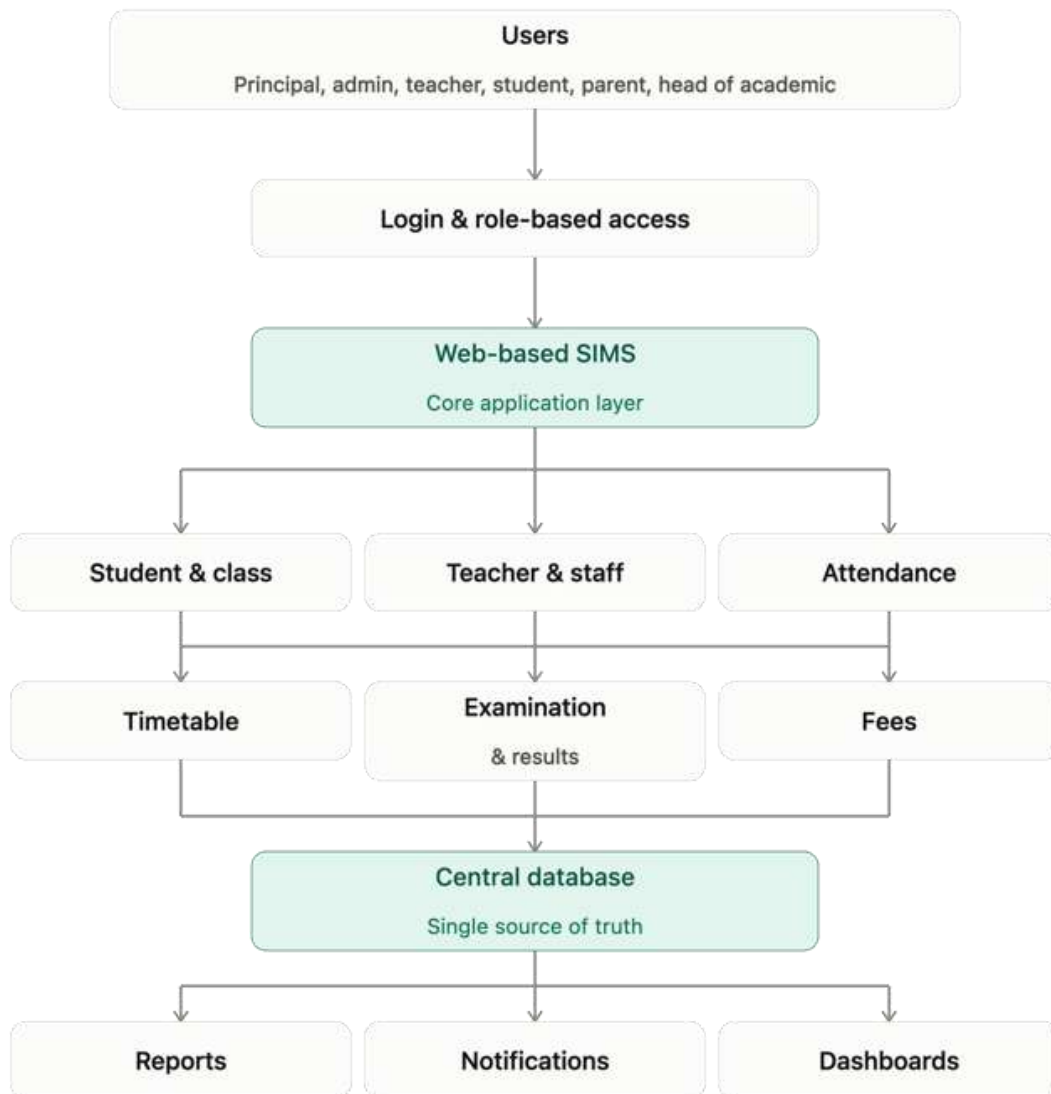
- Student and Class Management
- Teacher and Staff Management
- Student Attendance Management
- Timetable and Academic Scheduling
- Examination and Academic Performance Management
- Fee and Payment Management

In addition to these functions, the system includes supporting features such as user authentication, role-based dashboards, search functionality, report generation, announcements, and profile management. The application will operate as a Java-based web application using a centralized database accessible through a web browser.

2.7 Project Limitations

The first version of the system has several limitations. It will be developed as a Java-based web application within the project duration specified by the module. The system will not integrate with real online banking services or payment gateways and will not support biometric attendance devices. Additionally, a complete payroll management system is outside the scope of this project, with only basic teacher salary records being maintained. The system also depends on browser access, network availability, and accurate user input to ensure reliable operation.

System Overview Diagram



4. Functional Requirements

The proposed Web-Based School Information Management System will support six integrated major functional modules. Each module is designed to perform Create, Read, Update, and Delete (CRUD) operations, allowing authorized users to efficiently manage school information. Each major function is assigned to one group member for development and implementation.

4.1 Student & Class Management

This module manages student records and organizes students into their appropriate classes. It allows authorized users to register students, maintain personal and guardian information, assign students to classes, update enrolment details, and search student records.

4.2 Teacher & Staff Management

This module manages teacher profiles, staff information, academic assignments, attendance records, and basic salary information. It also supports assigning teachers to subjects and classes while maintaining accurate staff records.

4.3 Student Attendance Management

This module enables teachers to digitally record and monitor daily student attendance. It supports marking attendance, recording absence reasons, correcting attendance records, viewing attendance history, and generating attendance reports.

4.4 Timetable & Academic Scheduling

This module manages academic scheduling by creating and maintaining timetables. It supports assigning teachers, subjects, classrooms, and time slots while helping to organize class schedules efficiently.

4.5 Examination & Academic Performance

This module manages examinations, student marks, grading criteria, result publication, and academic performance reports. It also performs automatic grade calculations based on predefined grading criteria.

4.6 Fee & Payment Management

This module manages student fee structures, payment records, outstanding balances, receipt generation, and payment history. It enables administrators to maintain accurate financial records and summaries for the school.

All six modules provide CRUD (Create, Read, Update, and Delete/Deactivate/Cancel) functionality with role-based access control to ensure that only authorized users can perform specific operations.

5. Non-Functional Requirements

Non-functional requirements define the quality attributes and operational standards that the Web-Based School Information Management System must satisfy. These requirements ensure that the system is secure, efficient, reliable, easy to use, and maintainable while providing an optimal experience for all authorized users.

Quality Attributes	Requirements
Security	The system shall provide secure user authentication, role-based access control, and restricted data visibility to ensure that users can access only the information and functions authorized for their role.
Usability	The system shall provide a simple, user-friendly interface with clear navigation and readable layouts to enable school staff, teachers, students, and parents to use the system efficiently with minimal training.
Performance	The system shall support fast searching, record retrieval, and report generation to ensure efficient operation during daily school activities.
Reliability	The system shall maintain accurate data storage and ensure a stable connection with the centralized database to minimize data loss and system failures.
Maintainability	The system shall follow a modular design, allowing each major functional module to be developed, tested, updated, and maintained independently.
Privacy	The system shall protect sensitive information by allowing students and parents to view only their own authorized records and preventing unauthorized access to confidential data.

The above non-functional requirements ensure that the proposed system operates efficiently while maintaining security, reliability, and data privacy. They also contribute to the long-term sustainability of the application by supporting modular development and simplified maintenance.

6. Major Stakeholders / Users

The Web-Based School Information Management System is designed to serve multiple stakeholders within the school environment. Each stakeholder has specific responsibilities and is

provided with role-based access to perform only the functions relevant to their role. This ensures secure access to information while improving the efficiency of school operations.

User Role	Responsibilities
Principal	Views dashboards, reports, and overall school performance to monitor academic and administrative activities.
Administrator	Manages administrative records, student information, teacher records, fee management, and overall school operations.
Teacher	Marks student attendance, enters examination marks, manages assigned classes, and views academic information related to their teaching responsibilities.
Student	Views personal profile, class timetable, attendance records, examination results, and school notices through the system.
Parent	Monitors their child's attendance, examination results, fee payment status, and school announcements using authorized access.
Head of Academic	Manages academic planning, class allocations, timetable scheduling, subject assignments, and academic performance reports.

The implementation of role-based access control ensures that each stakeholder can access only the information and functions required for their responsibilities. This approach enhances system security, protects sensitive data, and promotes efficient collaboration among administrators, teachers, students, parents, and academic management.

7. Six Major Functions

The proposed Web-Based School Information Management System consists of six major functional approaches. Each function is designed to perform specific tasks that support the academic and administrative operations of the school. Every module includes Create, Read, Update, and Delete (CRUD) operations and is assigned to one group member for development. Role-based access control ensures that only authorized users can perform the functions relevant to their responsibilities.

7.1 Student & Class Management

Main Users: School Administrator, Head of Academic

Purpose

The Student & Class Management function is responsible for managing student records and organizing students into their appropriate classes. It provides a centralized repository for maintaining student information and class allocations, ensuring accurate and up-to-date records.

Expected Outcome

This module provides accurate and centralized student records with proper class allocation, making it easier to manage student information throughout the academic year.

Main Tasks

- Register new students.
- Maintain student personal and guardian information.
- Create and manage classes.
- Assign students to classes.
- Search and view student records.
- Update student enrolment status.

CRUD Operations

- Create: Register student, create class, assign student to class.
- Read: View student profiles and class lists.
- Update: Edit student information or change class assignment.
- Delete: Deactivate student or remove incorrect class assignment.

7.2 Teacher & Staff Management

Main Users: School Administrator, Head of Academic

Purpose

The Teacher & Staff Management function manages teacher profiles, academic assignments, and basic staff records. It helps maintain accurate information regarding teachers and their responsibilities within the school.

Expected Outcome

This function provides centralized teacher information with accurate subject and class assignments, supporting efficient staff management.

Main Tasks

- Register teachers.
- Manage teacher profiles and contact details.
- Assign teachers to subjects and classes.
- Record teacher attendance.
- Manage basic salary information.
- Search and view teacher records.

CRUD Operations

- Create: Register teacher, create assignment, record attendance.
- Read: View teacher profiles, assignments, attendance, and salary information.
- Update: Edit teacher information and assignments.
- Delete: Deactivate teacher or remove incorrect assignments.

7.3 Student Attendance Management

Main User: Teacher

Views: Student, Parent, Principal

Purpose

The Student Attendance Management function enables teachers to digitally record and monitor daily student attendance. It improves the accuracy of attendance records while providing easy access to attendance history and reports.

Expected Outcome

This function provides faster and more accurate attendance management with accessible attendance history and reporting capabilities.

Main Tasks

- Select assigned class and date.
- Mark students as Present, Absent, Late, or Excused.
- Record absence reasons.
- Correct attendance mistakes.
- View attendance history.

- Generate attendance reports.

CRUD Operations

- Create: Mark student attendance.
- Read: View daily attendance, attendance history, and reports.
- Update: Correct attendance information.
- Delete: Remove incorrect attendance entries.

7.4 Timetable & Academic Scheduling

Main User: Head of Academic

Views: Teacher, Student, Administrator

Purpose

This function manages academic scheduling by connecting classes, teachers, subjects, classrooms, and time periods into an organized timetable. It helps reduce scheduling conflicts and supports effective academic planning.

Expected Outcome

The system provides organized academic scheduling with clear teacher, subject, classroom, and time allocations.

Main Tasks

- Manage subjects and class schedules.
- Create timetable entries.
- Assign teachers, subjects, and rooms.
- Manage time slots.
- Update timetable changes.
- View class and teacher timetables.

CRUD Operations

- Create: Create timetable, subject, or schedule entry.
- Read: View class or teacher timetables.
- Update: Modify teacher, subject, room, or time allocations.
- Delete: Remove cancelled or incorrect timetable entries.

7.5 Examination & Academic Performance

Main Users: Teacher, Head of Academic

Purpose

The Examination & Academic Performance module manages examinations, student marks, grade calculations, result publication, and academic performance reports. The system automatically calculates grades based on predefined grading criteria.

Expected Outcome

The module provides accurate examination records, automatic grade calculations, and comprehensive academic performance analysis.

Main Tasks

- Create examinations.
- Enter student marks.
- Define grading criteria.
- Calculate grades automatically.
- Publish examination results.
- View academic performance reports.

CRUD Operations

- Create: Create examinations and enter student marks.
- Read: View examination results and academic performance reports.
- Update: Correct marks or examination details.
- Delete: Remove incorrect results or cancel examination records.

7.6 Fee & Payment Management

Main Users: School Administrator

Views: Parent, Principal

Purpose

The Fee & Payment Management function manages student fees, payment records, outstanding balances, and receipt generation. It provides accurate financial tracking while maintaining complete payment histories.

Expected Outcome

This function provides clear payment tracking with outstanding balances, payment receipts, and financial summaries to support school financial management.

Main Tasks

- Create fee types.
- Assign fees to students.
- Record payments.
- Calculate outstanding balances.
- Generate receipts.
- View payment history and financial summaries.

CRUD Operations

- Create: Create fee records and record payments.
- Read: View fees, payment history, and outstanding balances.
- Update: Correct fee or payment details.
- Delete: Cancel incorrect fee or payment records.

8. Minor Functions

The Minor Functions of the proposed Web-Based School Information Management System provide essential supporting features that enhance the functionality of the six major modules. Although these features are not considered standalone major modules, they improve system usability, security, navigation, and overall user experience. These functions are shared across the entire system and are accessible based on user roles and permissions.

Category	Features
Access & Security	Login, logout, password reset, role-based access control, and session management to ensure secure system access.
Navigation & Dashboard	Role-based dashboards, common navigation menus, and quick summaries that provide users with relevant information according to their role.
Search & Filtering	Search and filter functionality to quickly locate students, teachers, payments, examination results, and attendance records.
Reports	Generate attendance reports, examination result reports, fee reports, and management summaries to support decision-making.

Announcements	Publish school notices, examination updates, timetable changes, and reminders for students, teachers, and parents.
Profile Management	Allow users to view and update their permitted profile information while maintaining data privacy and security.

These supporting functions improve the efficiency and usability of the system by providing secure authentication, simplified navigation, quick access to information, comprehensive reporting, effective communication, and profile management for all authorized users.

9. System Limitations / Constraints

The proposed Web-Based School Information Management System has several limitations and constraints that define the scope of the first version of the project. These constraints help ensure that the system can be developed within the available time, resources, and assignment requirements while meeting its primary objectives.

The system will be developed as a Java-based web application, in accordance with the requirements of the Software Engineering module. The development period is limited to Week 3 through Week 14, requiring the project team to complete all analysis, design, implementation, testing, and documentation within the allocated timeframe.

The first version of the system will not support real online banking or payment gateway integration. Instead, fee payments will be recorded manually by the school administrator. Similarly, the system will not integrate with biometric or hardware-based attendance devices, and attendance will be recorded digitally through the web application by authorized teachers.

In addition, the system does not include a complete payroll management module. It will only maintain basic teacher salary information, while advanced payroll processing features such as tax calculations, allowances, and salary generation are beyond the scope of this project.

The application requires users to access the system through a web browser with a stable internet or local network connection. The accuracy and reliability of the information stored in the system depend on valid data entered by authorized users and the correct implementation of role-based access permissions.

Summary of System Constraints

- The system is developed as a Java-based web application.
- The project development period is limited from Week 3 to Week 14.
- Real online banking and payment gateway integration are not included in the first version.
- Biometric or hardware attendance device integration is not supported.

- Only basic teacher salary records are maintained; a complete payroll system is outside the project scope.
- The system requires browser access and an internet or local network connection.
- Data accuracy depends on valid user input and proper role-based access control.

10. Simple Timeline

The project will be completed over the period from Week 3 to Week 14, following a structured development schedule. The timeline includes the major activities required to successfully design, develop, test, and document the Web-Based School Information Management System. By dividing the work into weekly milestones, the project team can monitor progress, ensure timely completion of each phase, and effectively coordinate responsibilities among all group members.

Weeks	Planned Work
Week 3	Proposal presentation and proposal report preparation.
Week 4	Finalize system requirements, project scope, and role-based user permissions.
Week 5	Design the database, prepare the Entity Relationship Diagram (ERD), and develop the Use Case Diagram.
Week 6	Plan the user interface, implement authentication, and design the common system layout.
Week 7	Begin CRUD development for the six major functional modules.
Week 8	Continue module development and integrate the centralized database with the application.
Week 9	Integrate all modules, implement input validations, and conduct initial system testing.

Week 10	Prepare and deliver the progress presentation and submit the design document.
Week 11	Complete the remaining features, reports, and system improvements.
Week 12	Perform comprehensive testing, fix identified bugs, and improve the user interface.
Week 13	Conduct the final project presentation and viva examination.
Week 14	Finalize the project report, perform final improvements, and create system backups before submission.

The above timeline provides a clear roadmap for the project, ensuring that each stage of development is completed systematically. Following this schedule will help the team manage time effectively, monitor progress, and deliver the project within the allocated duration.

11. Conclusion

The proposed Web-Based School Information Management System offers a practical and effective solution for improving the management of academic and administrative activities within a school. By integrating major school operations into a single web-based platform, the system enables efficient management of student records, teacher information, attendance, timetables, examinations, and fee payments through a centralized database. This integration reduces the reliance on manual processes, minimizes data duplication, and improves the accuracy and accessibility of information.

The system is designed to support the needs of all key stakeholders, including the Principal, Administrator, Teachers, Students, Parents, and the Head of Academic. Through role-based access control, each user can securely access the information and functions relevant to their responsibilities. This enhances communication, improves decision-making, and supports the efficient day-to-day operation of the school.

Furthermore, the project has been structured into six major functional modules, allowing each group member to take responsibility for the design and implementation of a specific module. This modular approach encourages teamwork, simplifies system development, and makes testing and maintenance more manageable throughout the project lifecycle.

In conclusion, the proposed Web-Based School Information Management System is a feasible, valuable, and well-organized software solution that addresses the limitations of traditional school management methods. By following the planned development timeline and applying appropriate software engineering practices, the project team aims to deliver a secure, integrated, reliable, and user-friendly system that improves the overall efficiency and effectiveness of school management.

